The distributed nature of a search engine like Elasticsearch provides instant scalability that most databases are not designed for. A search engine developed with a backing database (with no full-text search capabilities) may not be able to provide *relevant* search results for queries, let alone cope with volumetric growth and serve results in real time.

> **Full-text searching with databases**
>
> Relational databases like Oracle and MySQL support full-text search functionality, albeit with less functionality than a modern full-text search engine like Elasticsearch. They are fundamentally different when it comes to storing and retrieving data, so you must be sure to prototype your requirements before choosing one of them. Usually, if the schemas are not going to change or data loads are low, and you already have a database engine with full-text search capabilities, beginning with full-text search on a database may make sense.

1.2.3 Databases vs. search engines

When building a search service on a traditional database, we need to consider and understand whether our requirements can be satisfied by the database efficiently and effectively. Most databases are designed to store large amounts of data but, unfortunately, are not well-suited for use as full-text search engines, for several reasons:

- *Indexing and search performance*—Full-text search requires efficient indexing and performant search and analytical capabilities, which traditional databases are not optimized for. Databases may struggle with indexing large volumes of data and, as a result, may exhibit poor query performance. Search engines like Elasticsearch and Solr are specifically designed to handle large amounts of text data and provide search results in near-real time. Search engines can handle large-scale data, indexing it and searching it much faster than traditional databases, as they are pretty much designed from the ground up for optimized search operations. Unfortunately, relational databases lack advanced search features such as fuzzy logic, stemming, synonyms, and so on.
- *Search*—Searching with traditional databases is more or less based on the exact matching of data values. While this is suitable for non-search-related find operations on structured data, it is a definite no-no for natural language queries, which are often complex. User queries are often misspelled, misconstructed grammatically, or incomplete and may contain synonyms and other language structures that databases fail to understand

  In natural language queries, users may not use the exact terms they are searching for (spelling mistakes), and unfortunately, traditional databases are not designed to support misspelled user input. This feature is supported by the fuzzy matching search function (words that are similar but not exactly the same) in modern search engines